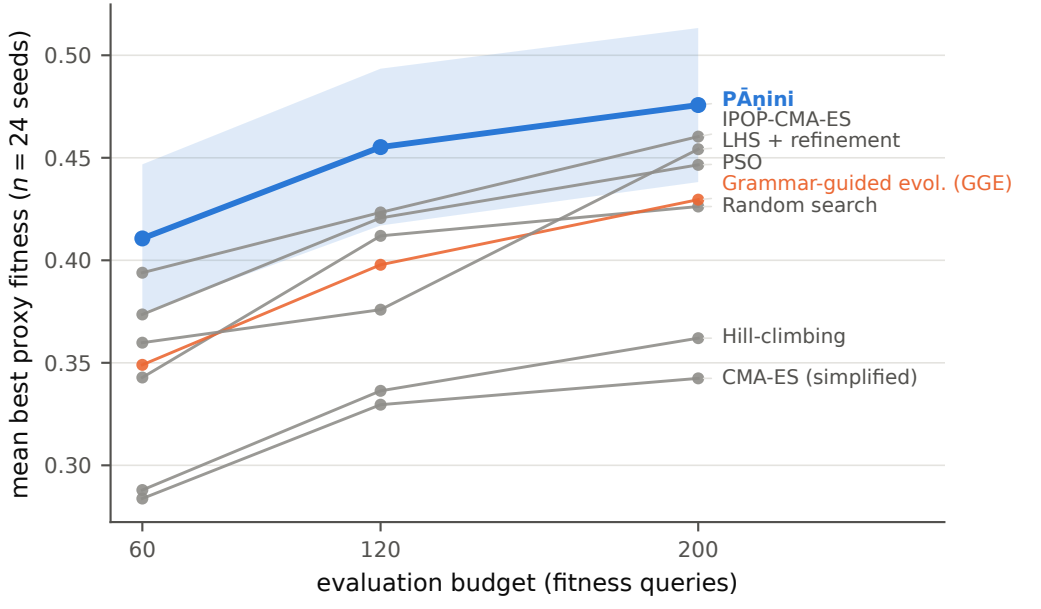
Budget sweep: Pāṇini (band: 95% CI) vs. GGE and six generic optimizers
mean best proxy fitness (n = 24 seeds)
0.50
0.45
0.40
0.35
0.30
60
120
200
evaluation budget (fitness queries)
Pāṇini
IPOP-CMA-ES
LHS + refinement
PSO
Grammar-guided evol. (GGE)
Random search
Hill-climbing
CMA-ES (simplified)